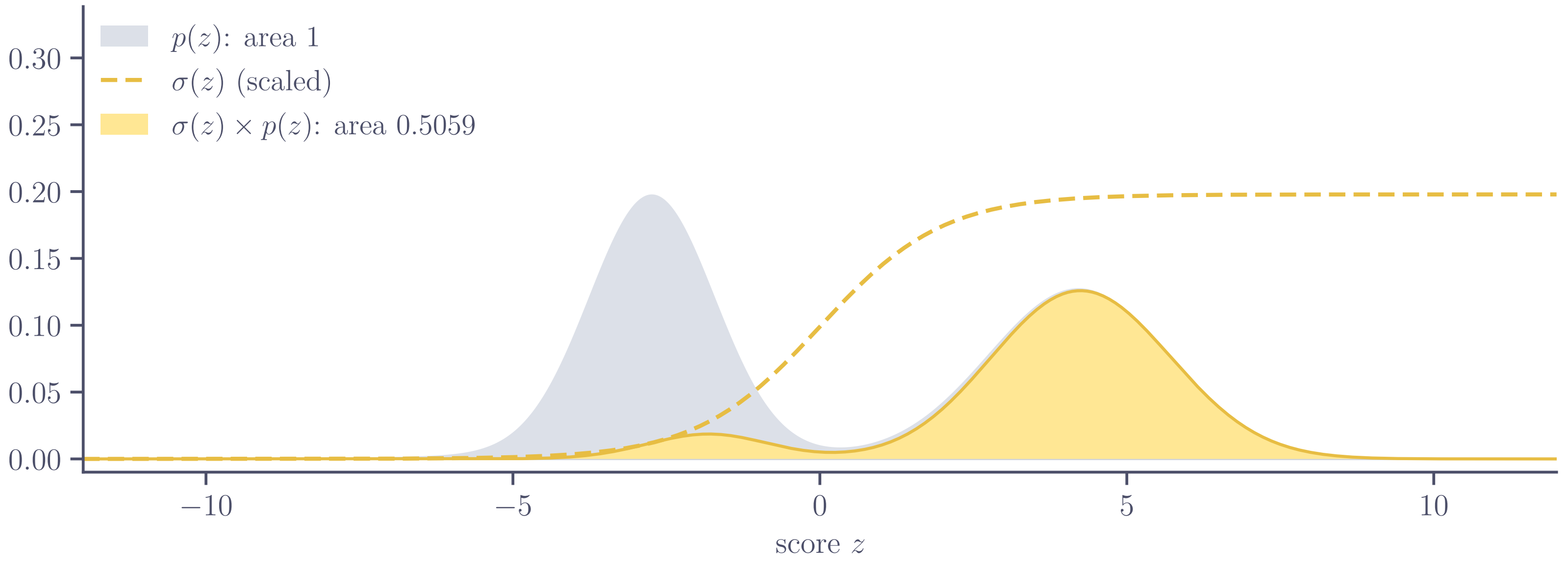
GOAL

Fit w on incomplete training points

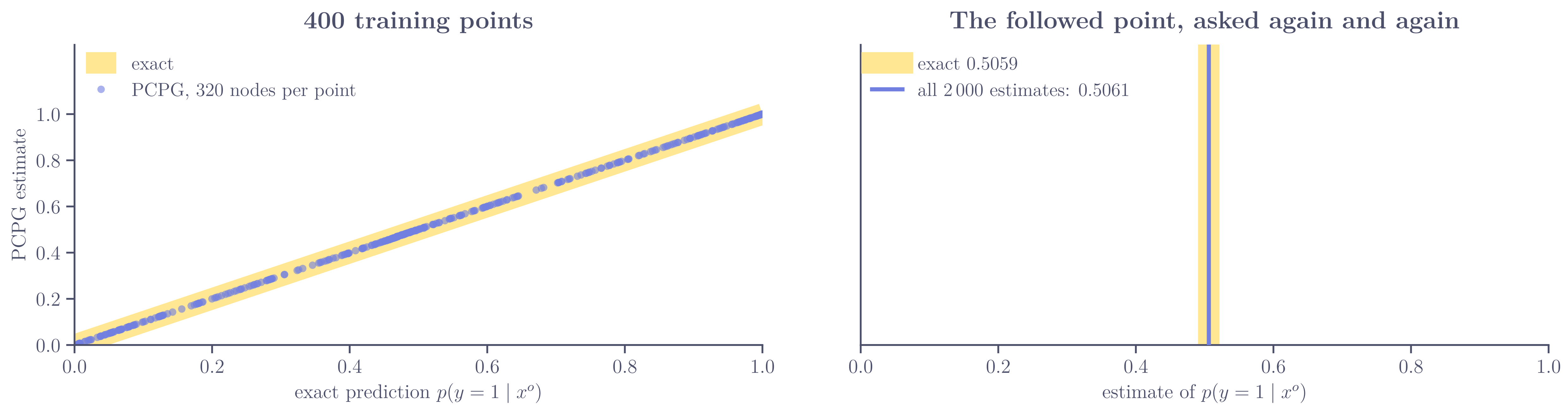
$$p(y=1\mid x^o) = \mathbb{E}_{p(z|x^o)}[\sigma(z)] = \int \sigma(z)\, p(z)\, dz$$



In training, also the training points have missing features (here 60% of them, at random). The density model is fit first, its likelihood leaves the missing features out exactly. Then w is fit to the prediction that is made at test time: for a point with observed x^o it is the yellow area under $\sigma(z) \times p(z \mid x^o)$, here for a training point with 9 of 15 features missing and label $y = 1$. Its loss is $-\log$ of the area of its label, and training minimises the mean loss over all points, $-\frac{1}{N}\sum_n \log p(y_n \mid x^o_n)$. So the area is needed for every point in every step, and the gradient of its log.

STEP 1

PCPG computes the prediction of every training point

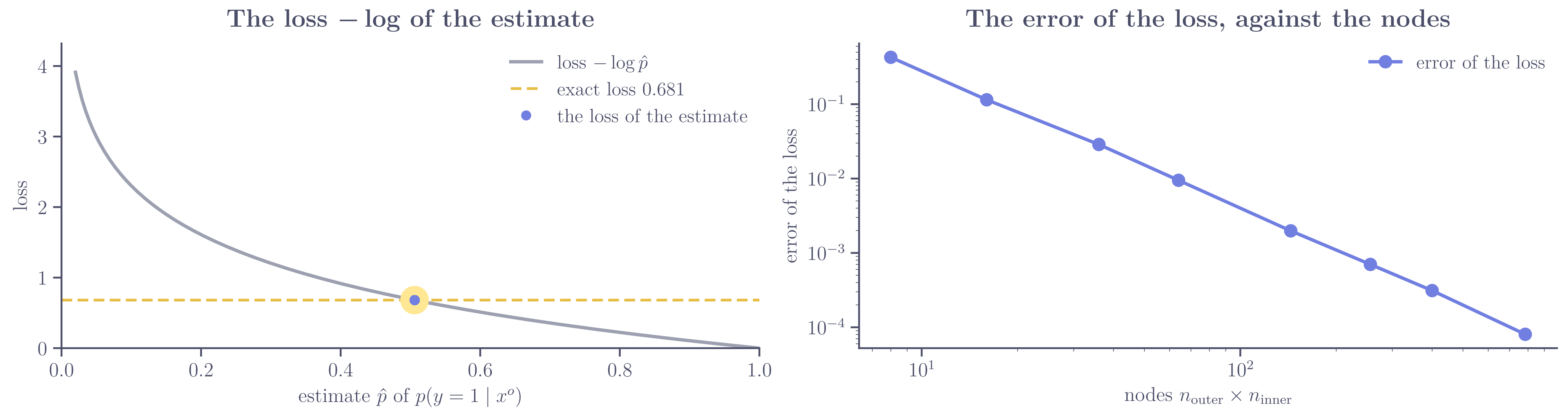

400 training points

The followed point, asked again and again

PCPG computes the area for every training point from the model's characteristic function (320 nodes per point). Every dot is one of 400 training points: its PCPG estimate against the exact value. They lie on the diagonal. The same point asked 2 000 times gives the same answer every time: PCPG is deterministic, a fixed function of w.

STEP 2

The loss of the estimate is the exact loss

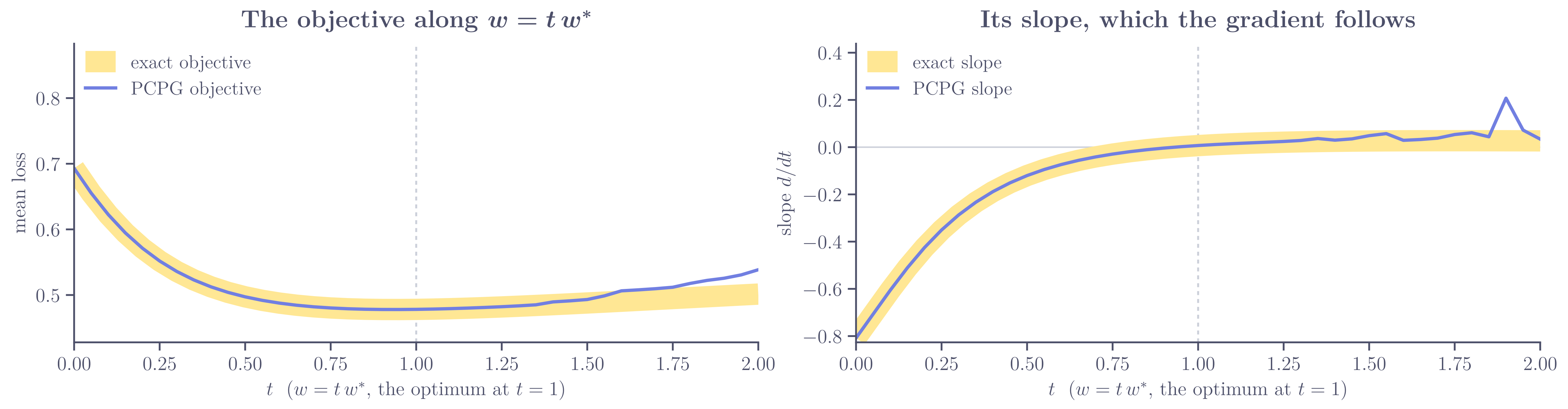

The loss $-\log$ of the estimate

The error of the loss, against the nodes

The loss is $-\log$ of the prediction of the label. As the prediction is exact up to the quadrature error, so is the loss: the dot sits on the curve, at the exact loss. The error of the loss falls quickly with the number of nodes $n_{\text{outer}} \times n_{\text{inner}}$, it is a deterministic error that does not average out, but it is small.

STEP 3

The objective and its slope are the exact ones

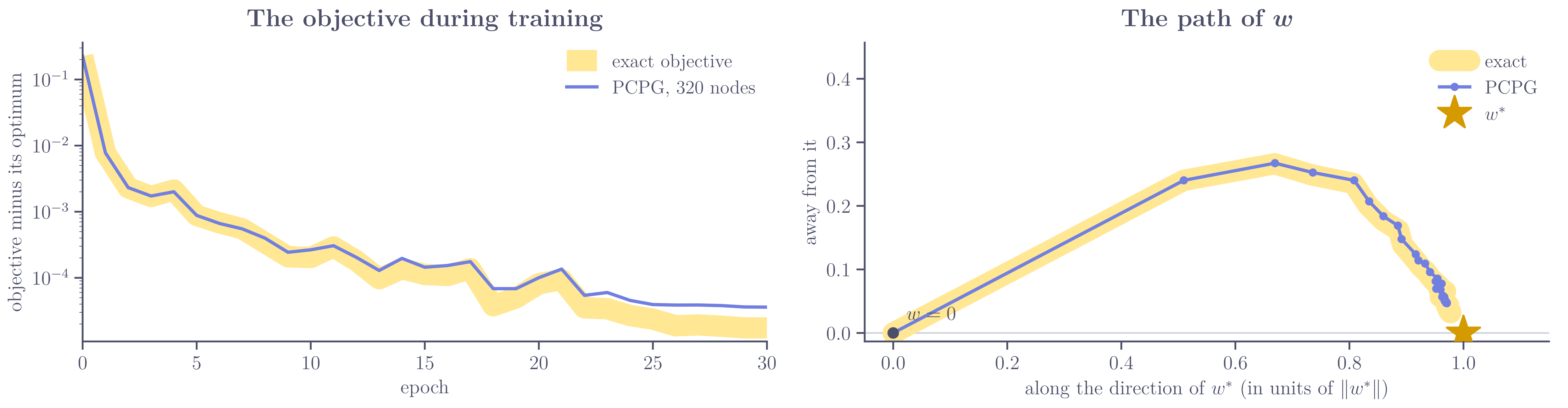

The objective along $w = t\,w^*$

Its slope, which the gradient follows

The mean loss over 2 000 training points along one direction of w, $w = t\,w^*$ (w^* the optimum of the exact objective, at $t = 1$). Around the optimum the PCPG objective is the exact one (in the yellow band), a smooth curve with its minimum at $t = 1$. For a much larger w the scores z get larger, the fixed nodes too coarse, and it drifts above. Its slope, which the gradient follows, is the exact one up to there, and the same at every call, so even full batch methods like L-BFGS can be used. Training starts at $w = 0$ and stays in the range where the nodes are good enough.

STEP 4

Training follows the exact objective


The objective during training

The path of w

Training with Adam on minibatches as in the experiments: the objective minus its optimum after every epoch. PCPG trains like the exact objective (yellow band) and ends at the same objective. The path of w from $w = 0$, along the direction of w^* and away from it (in units of $\|w^*\|$): the same as with the exact objective. Both stop a little short of w^* (star), where the objective is nearly flat.